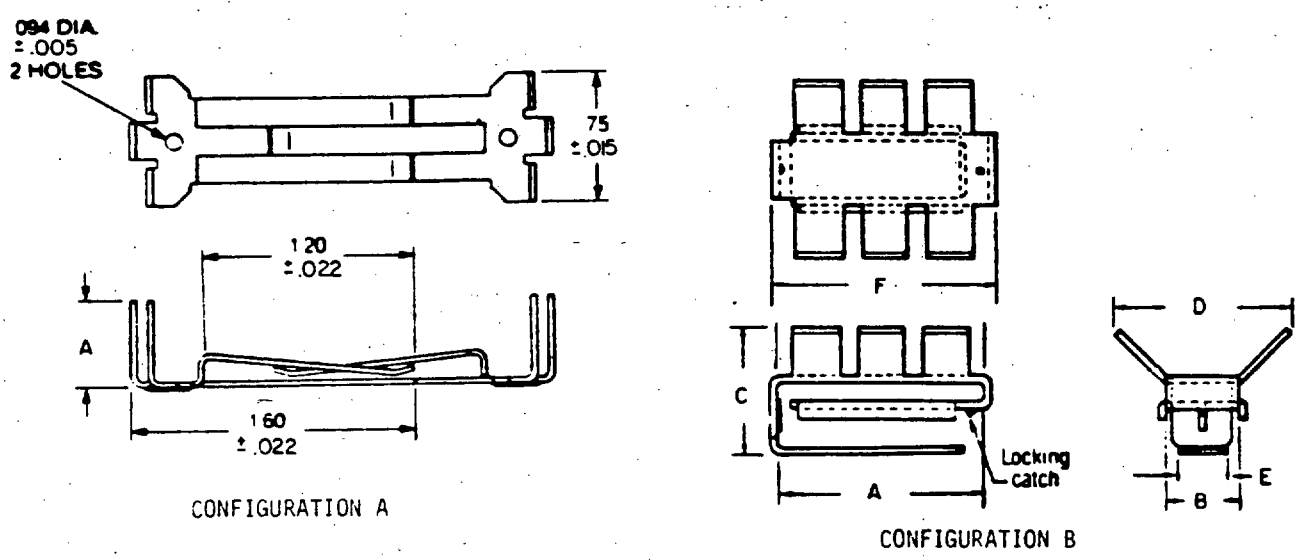
MILITARY SPECIFICATION SHEET

HEAT SINKS, SEMICONDUCTOR DEVICES, CLASS 1, (DUAL-IN-LINE PACKAGES)

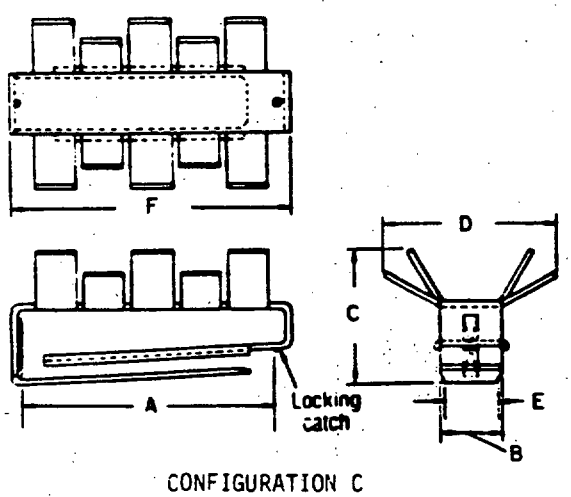
This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the heat sinks described herein shall consist of this specification sheet and the latest issue of MIL-H-87111.



CONFIGURATION A

CONFIGURATION B

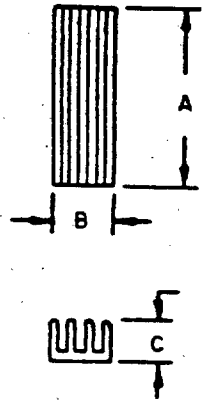


CONFIGURATION C

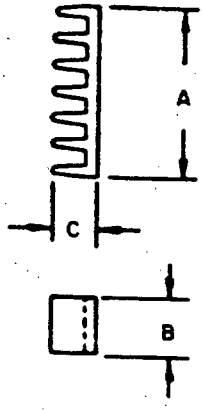
FIGURE 1. Heat sinks, (dual inline packages) - Continued.

FSC 5999

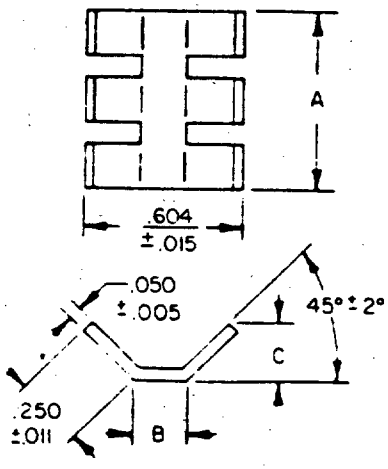
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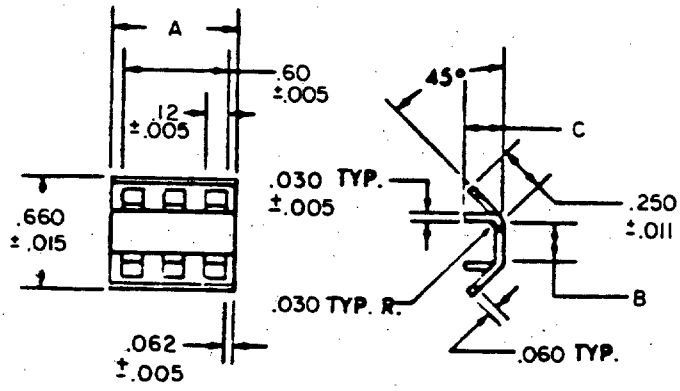
CONFIGURATION D



CONFIGURATION E



CONFIGURATION F



CONFIGURATION G

FIGURE 1. Heat sinks, (dual inline packages) - Continued.

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Dash Number M87111/07- 1/	IC Size (Pins)	Config-uration	Dimensions A
1B01	20-40	A	.250 (6.35)
1B02	20-40	A	.500 (12.70)
1B03	20-40	A	.750 (19.05)

1/ Complete dash number includes the letter B to indicate black anodized. Example of complete dash number: M87111/07-1B02.

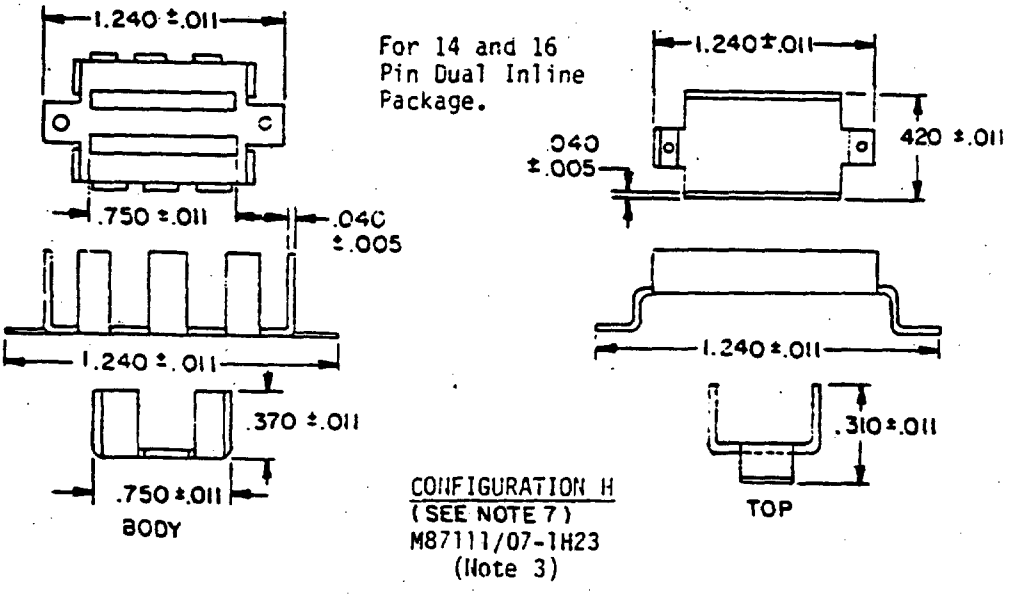
Dash Number M87111/07- (Note 4)	IC Size (Pin)	Config-uration	DIMENSIONS (Note 5)						Number of Fins
			A	B	C	D	E	F	
1-04	14/16	B	.837 (21.26)	.250 (6.35)	.410 (10.41)	.600 (15.24)	.200 (5.08)	.887 (22.53)	6
1-05	18	C	..930 (23.62)	.250 (6.35)	.556 (14.12)	.700 (17.78)	.210 (5.33)	1.125 (28.58	10
1-06	20	C	1.055 (26.80)	.250 (6.35)	.556 (14.12)	.700 (17.78)	.210 (5.33)	1.125 (37.08)	10
1-07	24	C	1.280 (32.51)	.540 (13.72)	.580 (14.73)	1.008 (25.60)	.480 (12.19)	1.460 (37.03)	12
1-08	40	C	2.090 (53.09)	.525 (13.34)	.590 (14.99)	.980 (24.89)	.480 (12.19)	2.256 (57.30)	20

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Complete dash number will include either B for black anodized or G for gold anodized. Example of complete dash number: M87111/07-1B08 or M87111/07-1G08.
5. Tolerances are ±.017 (0.43 mm) unless otherwise specified.

FIGURE 1. Heat sinks, (dual inline packages) - Continued.

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Dash Number M87111/07- (Note 3)	IC Size (Pin)	Config-uration (Note 4)	Dimensions			Number of Fins
			A (Note 5)	B ±.011 (0.28)	C ±.011 (0.28)	
1B09	14/16	D	.375 (9.53)	.250 (6.35)	.200 (5.08)	4
1B10	14/16	D	.750 (19.05)	.210 (5.33)	.240 (6.10)	3
1B11	14/16	D	.750 (19.05)	.250 (6.35)	.190 (4.83)	4
1B12	14/16	D	.750 (19.05)	.400 (10.16)	.240 (6.10)	5
1B13	24	D	1.250 (31.75)	.530 (13.46)	.190 (4.83)	5
1B14	28	D	1.450 (36.83)	.530 (13.46)	.190 (4.83)	5
1B15	40	D	2.000 (50.8)	.530 (13.46)	.190 (4.83)	5
1B16	64	D	3.200 (81.28)	.530 (13.46)	.190 (4.83)	5
1B17	14/16	E	.400 (10.16)	.250 (6.35)	.240 (6.10)	5
1B18	14/16	E	.730 (18.54)	.250 (6.35)	.190 (4.83)	7
1B19	14/16	E	.750 (19.05)	.250 (6.35)	.240 (6.10)	9
1B20	14/16	E	.750 (19.05)	.400 (10.16	.240 (6.10)	9
1-21	14/16	F	.750 (19.05)	.250 (6.35)	.212 (5.38)	NA
1B22	14/16	G	.750 (19.05)	.250 (6.35)	.230 (5.84)	NA

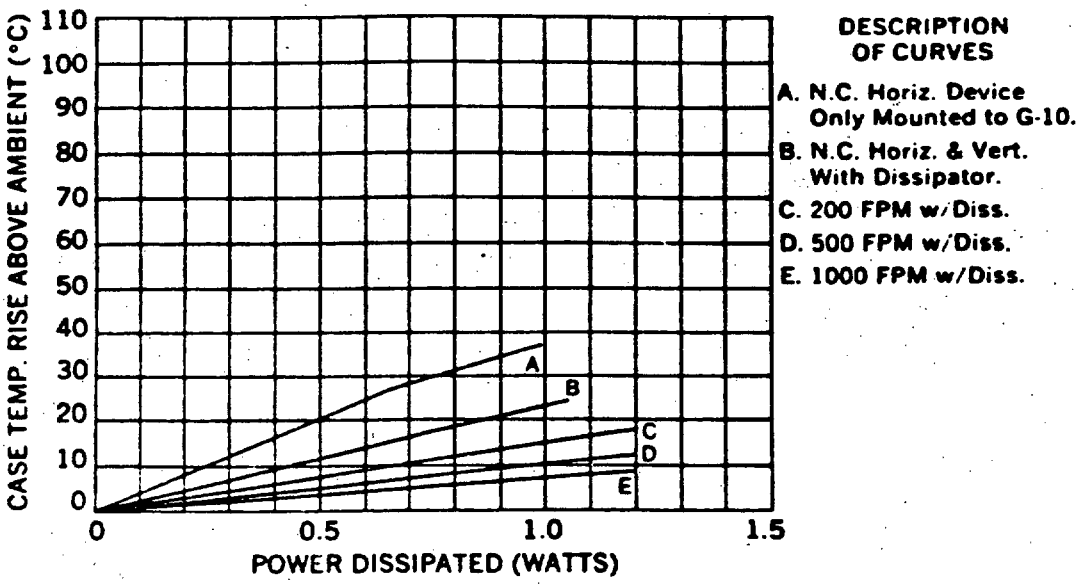


CONFIGURATION H
(SEE NOTE 7)
M87111/07-1H23
(Note 3)

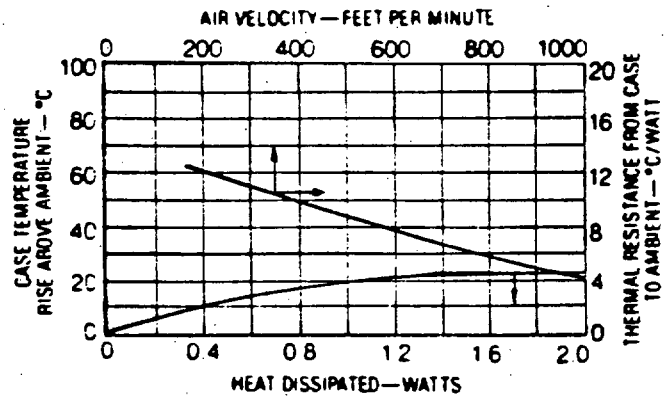
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Complete dash number includes the letter B to indicate part will have black anodized finish. Example of complete dash number: M87111/07-1B09.
4. Configuration F only: Dash number will include B for black anodized or R for gold chromate.
5. For lengths 1.000 (25.40 mm), tolerance is ±.015 (0.38 mm). For lengths over 1.000, tolerance is .022 (0.56 mm).
6. Metric equivalents are in parentheses.
7. Letter H in part number indicates hard anodized finish only.

FIGURE 1. Heat sinks, (dual inline packages) - Continued.

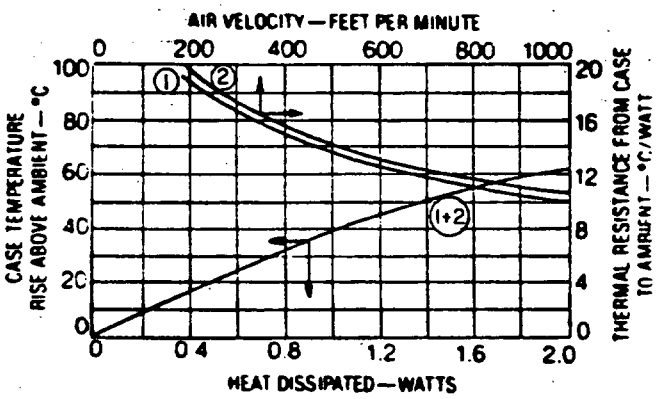
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CONFIGURATION A

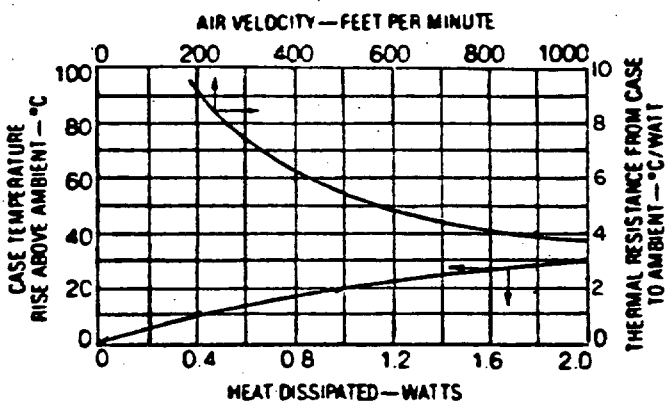


CONFIGURATION B



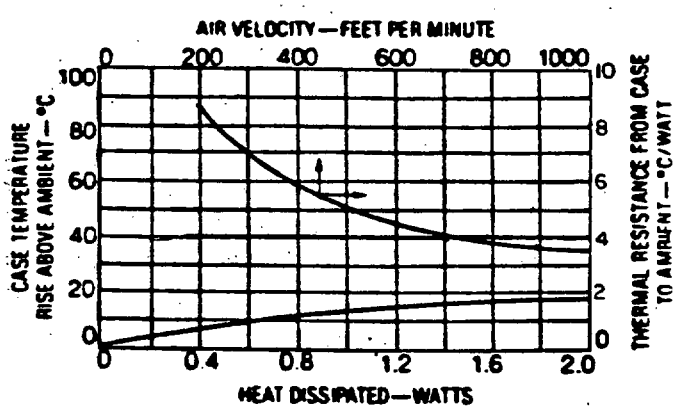
CONFIGURATION C

① M87111/07-1-05
② M87111/07-1-06



CONFIGURATION C

M87111/07-1-07

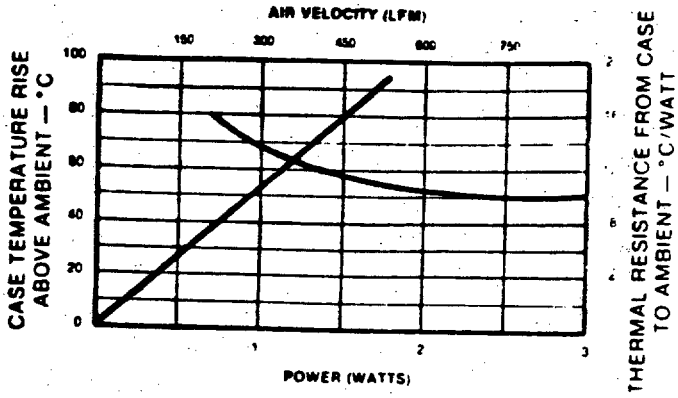


CONFIGURATION C

M87111/07-1-08

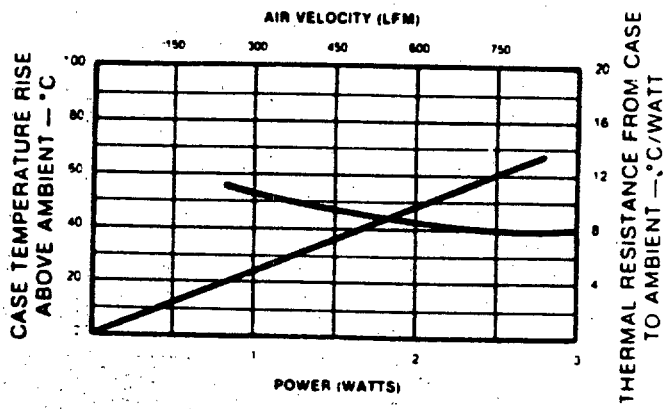
FIGURE 2. Thermal characteristics.

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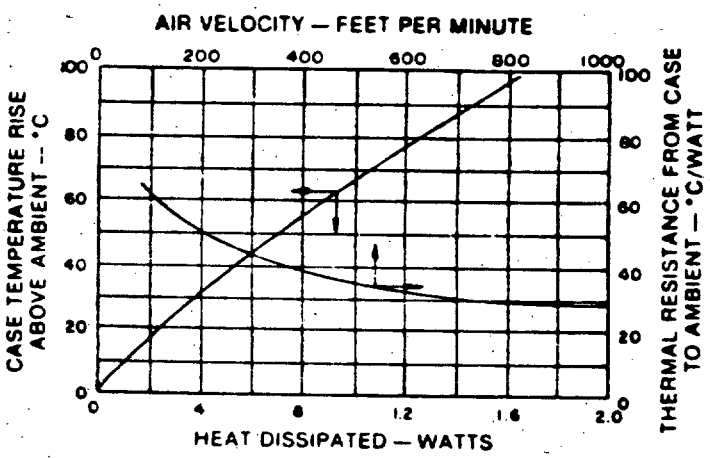
CONFIGURATION D

M87111/07-1B09



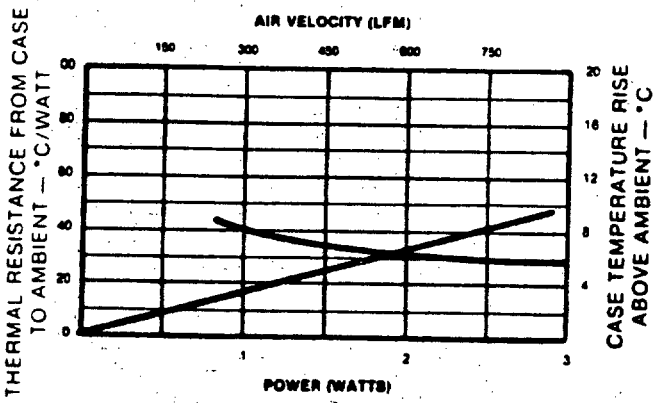
CONFIGURATION D

M87111/07-1B10



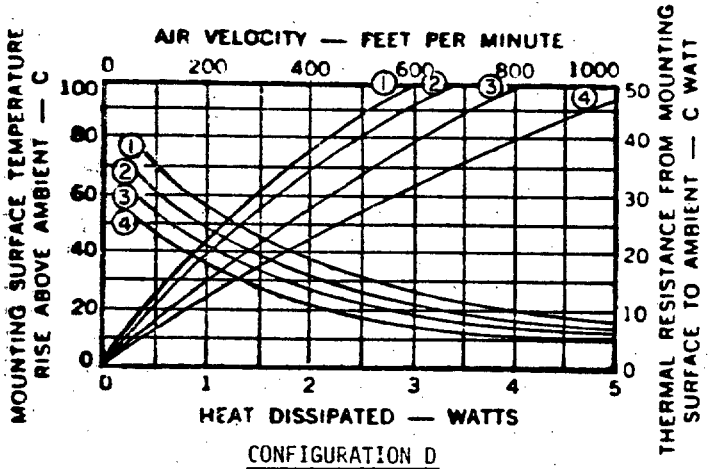
CONFIGURATION D

M87111/07-1B11



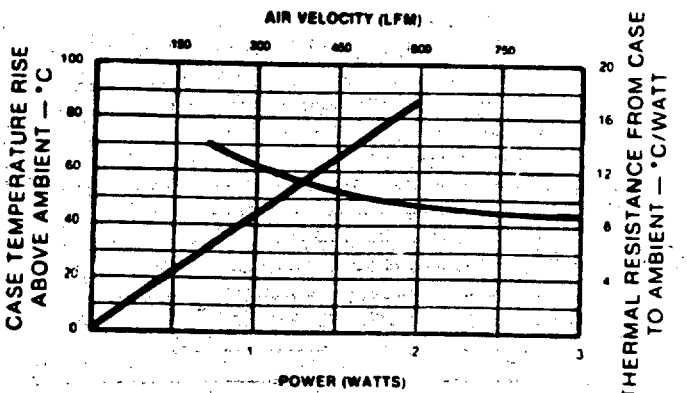
CONFIGURATION D

M87111/07-1B12



CONFIGURATION D

① M87111/07-1B13
② M87111/07-1B14
③ M87111/07-1B15
④ M87111/07-1B16

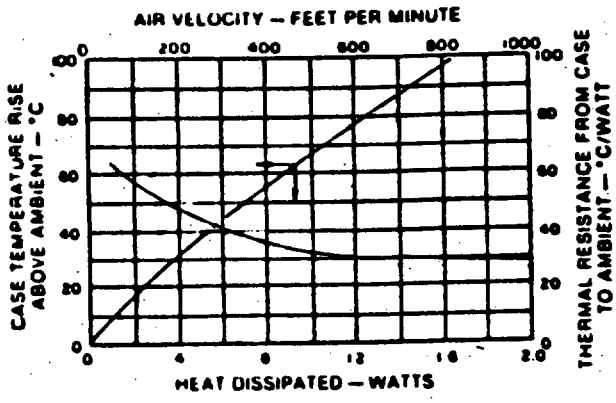


CONFIGURATION E

M87111/07-1B17

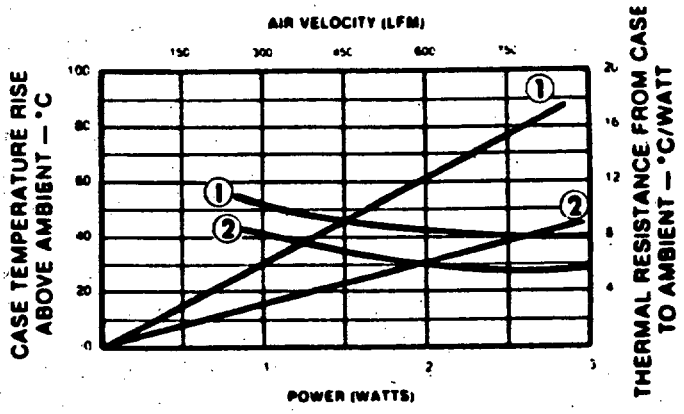
FIGURE 2. Thermal characteristics - Continued.

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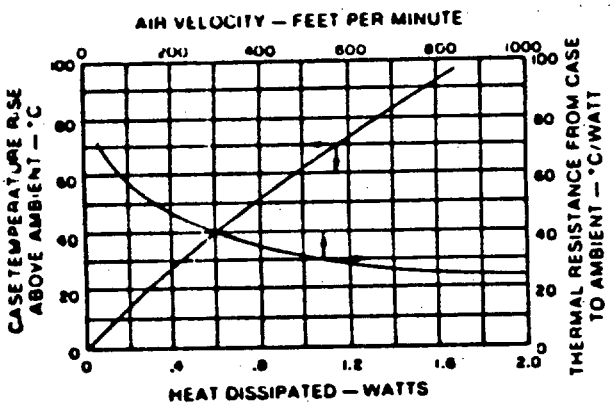
CONFIGURATION E

M87111/07-1B18



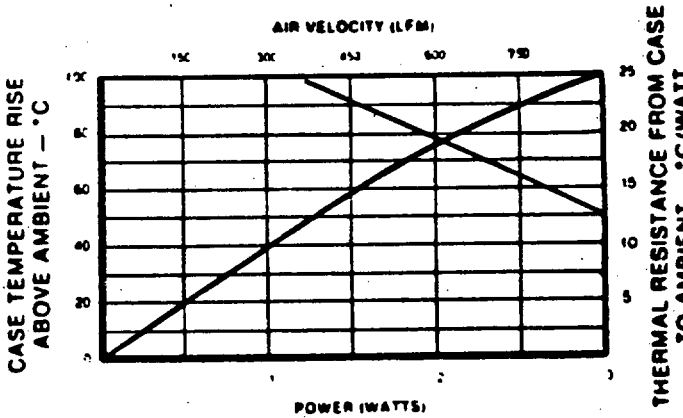
CONFIGURATION E

①M87111/07-1B19
②M87111/07-1B20



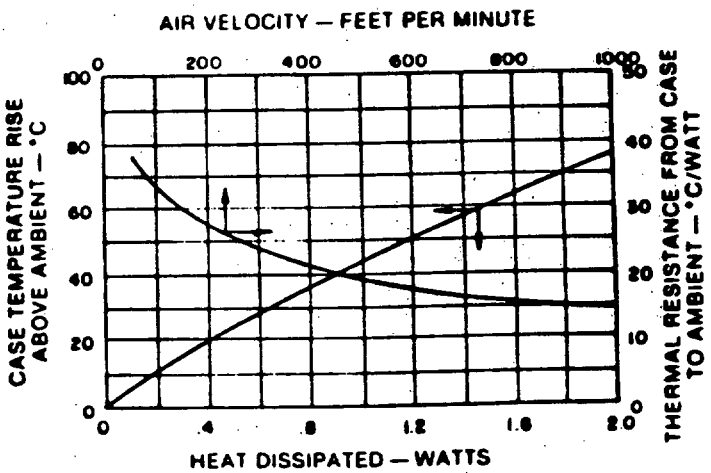
CONFIGURATION F

M87111/07-1-21



CONFIGURATION G

M87111/07-1B22



CONFIGURATION H

M87111/07-1H23

FIGURE 2. Thermal characteristics - Continued.

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REQUIREMENTS:

Design and configuration: See figure 1.

Material: Aluminum alloy.

Finish: Black anodize, gold anodize, hard anodize, gold chromate (B, G, H, or R, respectively).

Thermal resistance: See figure 2.

Mounting surface roughness: Not applicable.

Surface roughness: Not applicable.

JEDEC case size: Not applicable.

Fins: See figure 1.

Part number: M87111/07- (dash number from table I).

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - TD
Air Force - 11, 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Agent:
DLA - ES

Preparing activity:
Air Force - 85

(Project 5999-0139-7)

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